

Opportunity Dataset

Data Cleaning Documentation

Prepared by: Team 1

Program: Excelerate Data Visualization Trainee Program

Dataset: Opportunity Data Export (Copy_of__Opportunity_Data.xlsx)

WEEK 1 DELEVERABLE | 0308 SLU DVT TEAM 1

Active Team Members

Name	Role	Contribution
Queen Bella	Dataset Cleaning	Cleaned dataset, fee review, and cleaning log
Aman Gupta	Dataset Cleaning	Cleaned dataset and data quality review
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and Others	Supporting Contributors	Additional review and feedback across the team

1. Executive Summary

This document records the data cleaning process carried out on the Opportunity dataset supplied for the Data Visualization Trainee assignment. The raw export contained 5,733 records across 33 fields. Team 1 chose a non-destructive cleaning approach: rather than deleting rows suspected of being invalid, corrections were applied directly to formatting and structural issues, and anything uncertain; outlier values, suspected test records, incomplete key fields; was marked with a review flag instead of being removed. This keeps the full dataset intact and auditable, while still making problem records easy to isolate or exclude in downstream analysis.

Five rows were removed outright, as they contained corrupted, non-opportunity data (explained in Section 5.2). Every other record from the raw file is present in the cleaned output, now with consistent formatting and four flag columns indicating fee outliers, general quality concerns, and suspected test/placeholder records.

2. Objective and Scope

The objective was to take the raw Opportunity export and produce a dataset that is consistent, traceable, and safe to hand to a reviewer without hiding any judgment calls behind deleted rows. The scope covered:

- Structural review of the dataset; row count, column count, and data types.
- Identification and correction of formatting and quality issues, using the M.I.I.D.E. framework (Missing values, Incorrect entries, Inconsistent formatting, Duplicate/invalid records, Empty fields).

- Flagging, rather than removing any record where the team could not be fully certain a correction was appropriate.
- Production of three linked deliverables: the cleaned dataset (with flag columns and a cleaning log), a data dictionary, and this documentation.

3. Dataset Overview

The source file is a single-sheet export named `Copy_of__Opportunity_Data.xlsx`, containing one row per opportunity record (internships, courses, events, competitions, and related programs).

Attribute	Value
Source file	<code>Copy_of__Opportunity_Data.xlsx</code>
Raw row count	5,733
Raw column count	33
Primary key	<code>opportunity_id</code> (unique per record)
Rows removed (corrupted only)	5
Cleaned row count	5,728
Cleaned column count	36 (33 original + 3 flag columns)

4. Methodology

Cleaning was carried out in Python (pandas), using a repeatable script rather than manual spreadsheet edits, so that every transformation is documented and can be re-applied if the source data is refreshed. The process followed four stages: profiling the raw file to understand its structure and quality gaps; correcting what could be corrected with confidence (formatting, types, dates); and flagging rather than altering anything that required a judgment call the data owner should confirm.

5. Data Quality Issues and Corrective Actions

5.1 Missing Values

Twenty or more of the thirty-three fields had incomplete data. The most affected: `summary` (96.5% missing), `Testimonial` (85.2%), `role_responsibility` (81.2%), `Panellist` (79.5%), `current_editor` (96.2%), and `DropoutTransaction / NotStartedTransaction` (both above 99.7% missing). These were left as true blanks rather than filled with estimated values, since guessing would misrepresent the source data. Fields that are almost entirely empty are called out in Section 8 for follow-up with the data owner, since removing a column is a schema decision, not a cleaning one.

5.2 Incorrect and Invalid Records

Five records were removed outright because they did not represent real opportunity data. Three contained JSON code fragments in place of normal values for example, `%22label%22: %22Yemen%22` sitting inside the `category` and `currency_type` fields, a sign of a row misalignment during the original export. The remaining two had an `opportunity_id` prefixed "Testimonial#" instead of "Opportunity#", meaning they were a different record type entirely that had leaked into this file. In every case, no field in the row could be reliably repaired, so removal was the only defensible option.

5.3 Inconsistent Formatting

Several categorical fields used inconsistent labels for the same underlying value. The location field showed at least nine variants for what should be two consolidated categories — case differences ("Virtual"/"virtual"), a typo ("vitruual"), and abbreviations ("WFM", "wfm", "Work From Home", "WORK FROM HOME") all merged into Virtual or Work From Home. The currency_type field listed EUR and EURO separately; these were merged into a single EUR label. The duration_type field mixed singular and plural units and contained a typo ("yearssss"); all units were standardized to plural form. The is_archived and is_auto_approve fields used inconsistent True/False representations and were normalized to a simple 1/0 flag. The three date fields (created_at, modified_at, last_date_to_apply) were stored as raw epoch-millisecond numbers and converted to calendar dates.

5.4 Suspected Test, Demo, and Placeholder Records

A review of the name and location fields identified 4,186 records that show strong signs of being internal test, demo, or automation-generated entries rather than genuine opportunity listings — names such as "Course Automation 1740727622332" and "Digital Marketing - Demo," and Lorem-ipsum placeholder text in the location field such as "qwerty" or "Elit cum quis volup."

These records make up a large share of the raw file over 70% and would distort any category-level or trend analysis if treated as genuine data. Rather than deleting them, the team flagged every matching row with is_test_or_placeholder = "Yes - likely test/demo record." This keeps the decision reversible: an analyst can filter these rows out for a production-ready view, or include them if the review confirms they are not, in fact, test data. Deleting on a name-pattern assumption was judged too high-risk for a shared team deliverable, since it cannot be undone without returning to the raw export.

5.5 Fee and Data Quality Outliers

The fee field contained values ranging from 0 up to unrealistic outliers. Rather than applying a single global threshold, fee values were reviewed relative to their currency, and 1,103 records were marked in the fee_flag column as "Review - unusually high fee." A broader quality_review_flag column also captures secondary issues found during row-level review most notably 331 records where the fee currency was missing, and 24 records where the application deadline (last_date_to_apply) fell before the record's own creation date (created_at), which is logically inconsistent and worth a second look.

5.6 Empty and Incomplete Fields

Separately from true missing values, a number of cells across the dataset contained the literal word "Null" typed as text rather than a genuine empty cell, and leading/trailing whitespace was present in several text fields. Both were corrected: literal "Null" text was converted to a true blank, and whitespace was trimmed throughout.

6. Summary of Actions Taken

Issue	Action Taken	Rows / Cells Affected
Corrupted or invalid records	Removed	5
location inconsistency	Standardized to 2 core categories	~11 raw variants merged
currency_type inconsistency	EUR/EURO merged	334 rows affected
duration_type inconsistency	Standardized to plural form	Dataset-wide
Boolean field inconsistency	Normalized to 1/0	is_archived, is_auto_approve
Date fields as raw numbers	Converted epoch ms to calendar date	created_at, modified_at, last_date_to_apply
Suspected test/demo records	Flagged, not deleted	4,186

Issue	Action Taken	Rows / Cells Affected
Fee outliers	Flagged for review	1,103
Secondary quality issues	Flagged (missing currency, deadline logic)	1,467
Literal "Null" text / whitespace	Corrected	Dataset-wide

7. Validation

After cleaning, the output was re-checked to confirm: opportunity_id remains unique across all 5,728 retained rows; all three date fields parsed with no invalid values; location, currency_type, and duration_type show only the expected standardized values; and every flag column (fee_flag, quality_review_flag, is_test_or_placeholder) is either blank or a recognized flag value, with no partial or malformed entries.

8. Limitations and Recommendations

- Several fields (DropoutTransaction, NotStartedTransaction, CareerAddOn, current_editor) are populated in fewer than 8% of records. Team 1 recommends confirming with the data owner whether these fields are still actively collected before considering their removal in a future schema review.
- The is_test_or_placeholder flag is based on naming and location patterns, not a confirmed system source field. It should be treated as a strong signal for review, not an automatic exclusion rule.
- The 24 records where the deadline precedes the creation date should be checked against the source system, since this points to either a data entry error or a timestamp field being populated incorrectly upstream.
- The tracking_questions field contains nested JSON metadata and was left unchanged; a separate parsing step would be needed if its contents are required for analysis.

9. Deliverables

This documentation accompanies two data files, both produced from the same cleaning approach described above:

Deliverable	Description	Link
Team1_Cleaned_Opportunity_Dataset.xlsx	Cleaned dataset (5,728 records, 36 columns) with Cleaning Log and Fee Review sheets	Open spreadsheet
Team1_Data_Dictionary.xlsx	Column-by-column reference guide for all fields, including flag columns	Open spreadsheet

10. Conclusion

The Opportunity dataset is now consistently formatted, fully traceable, and free of only the genuinely unusable records. Every other judgment-call-suspected test data, fee outliers, and secondary quality concerns have been preserved in the dataset and clearly flagged rather than silently removed. Team 1 considers this the safer and more transparent standard for a shared deliverable, and recommends it be treated as the team's final submission for this stage of the trainee assignment.